Linear-Error Clique Partitions of Split Graphs via Structured Triangle Packing

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Abstract

Let $G = (K \cup I, E)$ be a split graph on n vertices, with K a clique and I an independent set. We prove that there is an absolute constant C such that

$$|E(G)| - 2\nu_3(G) \le \frac{n^2}{6} + Cn,$$

where $\nu_3(G)$ is the maximum number of pairwise edge-disjoint triangles in G. Consequently,

$$\mathrm{cp}(G) \le \frac{n^2}{6} + Cn$$

for every split graph G, where $\mathrm{cp}(G)$ is the minimum number of cliques whose edge sets partition $E(G)$. Chen, Erdős, and Ordman proved the earlier bound $3n^2/16 + O(n)$ for split graphs [5]. The present result improves its quadratic coefficient to the sharp value $1/6$, matching the complete-split lower-bound family and establishing the $n^2/6 + O(n)$ scale for split graphs. Subsequent chordal bounds appear in Paper IV [IV]. The exact family has value $n^2/6 + n/6$, so a linear term is necessary; we do not determine the least uniform linear coefficient.

The proof separates three regimes according to $\alpha = |I|/|K|$. In the bulk regime, an exact four-orbit linear program for a common-neighborhood profile yields a uniform quadratic fractional margin; the Haxell–Rödl theorem then absorbs the subquadratic integrality loss. When $\alpha \to 0$, large edge-disjoint matchings anchored at the independent vertices leave an almost complete, triangle-divisible clique residual, which is decomposed exactly by dense graph decomposition results. When $\alpha \to 2$, averaged factorization closes the short corridor, while a double-factor polarization inequality and a shifted-center gain-completion argument close the remaining mesoscopic corridor.

The proof does not establish the stronger universal estimate $\nu_3^*(G) - \nu_3(G) = O(n)$. Away from the extremal corridor, a potentially superlinear integrality gap is absorbed by quadratic fractional slack.

All corridor-specific mechanisms are proved in the paper; in particular, the list edge coloring ingredient is established from first principles in Appendix D.

Keywords: clique partition; split graph; triangle packing; fractional packing; factorization; polarization; list edge coloring; graph decomposition.

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1. Introduction

1.1 Clique partitions of split graphs

A **clique partition** of a graph G is a family of complete subgraphs whose edge sets partition $E(G)$. Cliques of order two are allowed. The minimum size of such a family is denoted by $\mathrm{cp}(G)$.

The chordal clique-partition problem of Erdős, Ordman, and Zalcstein asks whether every chordal graph satisfies $\mathrm{cp}(G) \le n^2/6 + O(n)$ [8]. Their complete-split (indeed threshold) construction shows that the coefficient $1/6$ is unavoidable. Split graphs form a natural subclass of chordal graphs: a graph is split if its vertex set can be partitioned into a clique and an independent set. Chen, Erdős, and Ordman proved that every split graph satisfies $\mathrm{cp}(G) \le 3n^2/16 + O(n)$ [5]. Corollary 1.2 improves the quadratic coefficient from $3/16$ to the sharp value $1/6$, matching the lower bound at quadratic order and establishing the conjectured linear-error scale for split graphs. Paper IV subsequently establishes the chordal bound and structural stability results [IV].

This is the third paper in the series on the Erdős–Ordman–Zalcstein clique-partition problem (Erdős Problem #81 [3]). Paper I proves the finite fractional split-graph bound $|E(G)| - 2\nu_3^*(G) \le n^2/6 + n/2$ [15]. Paper II identifies the complete-split terminal family and determines the exact finite maximum of the fractional cover functional over chordal graphs [16]. Papers I and II concern fractional optimization. Paper IV addresses chordal bounds and structural stability [IV]; Paper V develops the general transfer framework [V]. Neither later paper is an input to this proof. The present paper is the first integral installment: it strengthens the split-graph estimate to the linear-error scale $n^2/6 + O(n)$.

Series notation. Paper I writes the fractional deficit as $\Phi^*(G) = |E(G)| - 2\nu_3^*(G)$, Paper II writes the equal cover-side quantity as $\Phi_\tau(G)$, and the present paper uses the unstarred $\Phi(G) = |E(G)| - 2\nu_3(G)$ for the integral deficit.

Paper II of the series determines the fractional extremal problem exactly. For every n,

$$\max_{\substack{|V(G)|=n \\ G \text{ chordal}}} \big(|E(G)| - 2\nu_3^*(G)\big) = \max_{\substack{|V(G)|=n \\ G \text{ split}}} \big(|E(G)| - 2\nu_3^*(G)\big) = \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor.$$

It follows from the dense-graph fractional-to-integral packing theorems of Haxell–Rödl and Yuster [11,17] that every chordal graph satisfies

$$\mathrm{cp}(G) \le \left(\frac{1}{6} + o(1)\right) n^2.$$

The present paper proves the second-order refinement $n^2/6 + O(n)$ for arbitrary split graphs.

1.2 Triangle packings and the linear-error target

Let $\nu_3(G)$ be the maximum number of pairwise edge-disjoint triangles in G. Every such packing gives a clique partition: keep every packed triangle and use one K_2 for each uncovered edge. Hence

$$\mathrm{cp}(G) \le |E(G)| - 2\nu_3(G). \tag{1.1}$$

The central theorem of this paper is an integral estimate at the correct second-order scale.

Theorem 1.1 — Linear-error split-graph theorem

There exists an absolute constant C such that every split graph G on n vertices satisfies

$$\boxed{|E(G)| - 2\nu_3(G) \le \frac{n^2}{6} + Cn}. \tag{1.2}$$

Combining (1.1) and Theorem 1.1 gives the clique-partition statement.

Corollary 1.2 — Linear-error clique partition bound

There exists an absolute constant C such that every split graph G on n vertices satisfies

$$\boxed{\mathrm{cp}(G) \le \frac{n^2}{6} + Cn}. \tag{1.3}$$

Section 10.2 evaluates the Erdős–Ordman–Zalcstein complete-split family $K_p \vee \overline{K}_{2p}$ exactly. It separates the sharp quadratic coefficient $1/6$ from the still-undetermined least uniform linear coefficient.

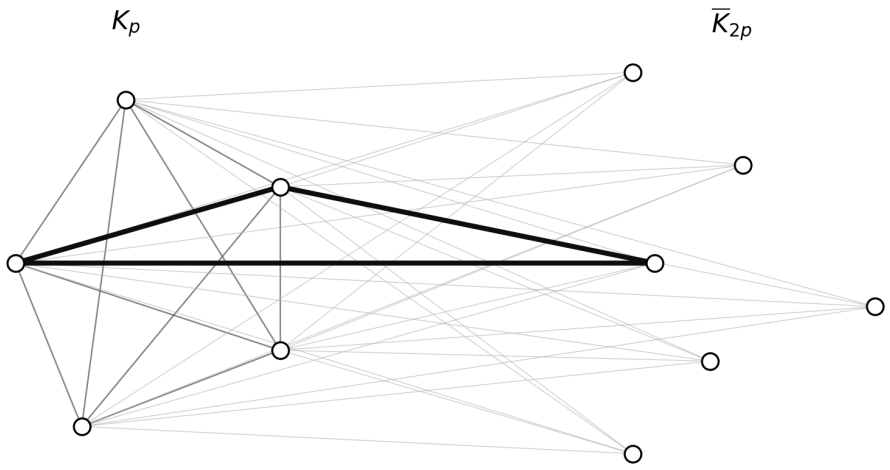


Figure 1: The complete-split graph $K_p \vee \overline{K}_{2p}$. Every edge between the two parts is present, and the independent side has no internal edges. Since every triangle contains a clique edge, an edge-disjoint triangle packing uses at most $\binom{p}{2}$ triangles; see Section 10.2. The figure is illustrative and is not used as a premise of the proof.

1.3 Why the mesoscopic corridor appears

The localization argument is bookkeeping inside the contradiction proof, not an additional extremal proposition. Suppose that no absolute linear-error bound holds, and for each positive integer k choose a minimum-order split graph G_k satisfying

$$|E(G_k)| - 2\nu_3(G_k) > \frac{|V(G_k)|^2}{6} + k|V(G_k)|.$$

After passing to a subsequence, writing $|K_k| = p_k$, $|I_k| = q_k$, every such hypothetical sequence must satisfy

$$\frac{q_k}{p_k} \longrightarrow 2.$$

Moreover, if $q_k = 2p_k - s_k$, then the bulk and short-corridor estimates force

$$\sqrt{p_k} \ll s_k = o(p_k).$$

Thus the proof is forced into the mesoscopic near-extremal corridor. The high/low-dispersion dichotomy in Section 9 then excludes that corridor. Section 10.3 returns to this localization only to explain the architecture of the completed proof.

1.4 What the theorem does not prove

Let $\nu_3^*(G)$ denote the fractional triangle-packing number. The present proof does **not** establish

$$\nu_3^*(G) - \nu_3(G) = O(n) \tag{1.4}$$

uniformly over split graphs.

Indeed, write

$$\Gamma(G) = \nu_3^*(G) - \nu_3(G)$$

and

$$S(G) = \frac{n^2}{6} - \big(|E(G)| - 2\nu_3^*(G)\big).$$

Then

$$|E(G)| - 2\nu_3(G) = \frac{n^2}{6} - S(G) + 2\Gamma(G). \tag{1.5}$$

Theorem 1.1 only requires

$$2\Gamma(G) \le S(G) + O(n).$$

In the bulk regime, $S(G)$ is quadratic and absorbs the general $o(n^2)$ integrality loss. Thus the universal linear integrality-gap problem remains open.

1.5 Proof architecture

Write

$$|K| = p, \qquad |I| = q, \qquad \alpha = \frac{q}{p}.$$

The ratio $\alpha = q/p$ determines which source of slack is available. We first dispose of $q \geq 2p - 1$ by direct averaged factorization, and may then assume $0 \leq \alpha < 2$. A hypothetical sequence of counterexamples has a subsequence in one of three regimes.

1. **Bulk:** α stays away from both 0 and 2. An exact common-profile LP and a fractional cloning argument produce quadratic fractional slack. Haxell–Rödl converts the fractional packing into an integral one with only $o(n^2)$ loss.
2. **Sparse independent side:** $\alpha \to 0$. Each independent vertex anchors a large matching inside its neighborhood. The residual clique is almost complete. After deleting $O(p)$ edges to correct divisibility, it admits an exact triangle decomposition.
3. **Near-extremal corridor:** $\alpha \to 2$. Write $q = 2p - s$. Averaged factorization closes $s = O(\sqrt{p})$. For $\sqrt{p} \ll s = o(p)$, high dispersion is paid by a double-factor polarization term, while low dispersion implies closeness to a common center and is handled by shifted-center gain completion.

Table 1. The three proof regimes. The bulk and sparse regimes retain their literature provenance, while the near-extremal corridor is elementary and effective.

Regime	Condition on $\alpha = q/p$	Source of slack	Mathematical input	Sections
Sparse independent side	$\alpha \to 0$	dense clique residual admits an exact triangle decomposition	Dross + Barber–Kühn–Lo–Osthus	§8
Bulk	α bounded away from 0 and 2	quadratic fractional margin (common-profile LP + fractional cloning)	Haxell–Rödl/Yuster	§3–4, §9.1
Near-extremal corridor	$\alpha \to 2$, $q = 2p - s$	averaged factorization, double-factor polarization, shifted-center gain completion	none beyond standard foundations	§5–7, §9.2–9.3, §10.5

The division is not cosmetic. In the bulk there is enough quadratic fractional slack to absorb a general $o(n^2)$ rounding loss. At the two ends that slack vanishes, so the proof replaces generic rounding by constructions adapted to the geometry of the split presentation.

1.6 Supplementary computational audits

The accompanying closure package contains exact-arithmetic and finite-instance regression tests for selected identities and quantitative inequalities used in the proof.

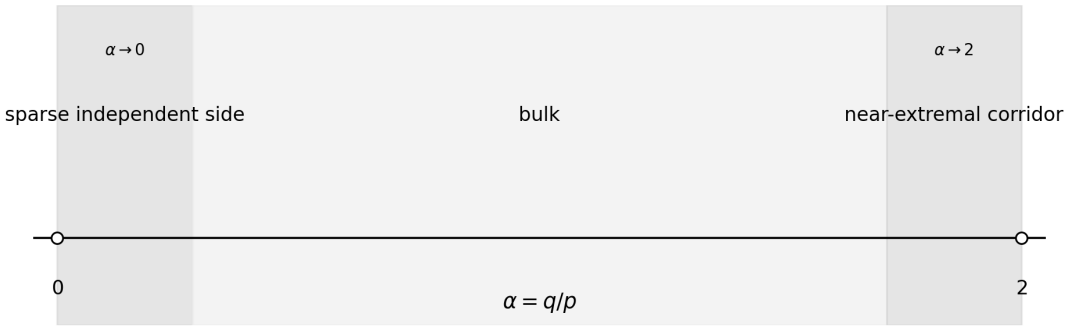


Figure 2: The three parameter regimes used in the proof. The principal corridor thresholds are shown; the complete hypotheses are stated in Sections 4, 8, and 9. The endpoint regimes are asymptotic rather than hard subintervals of $[0, 2]$.

Every audited statement is proved analytically in the manuscript. No computation, script, finite enumeration, or solver output is a logical premise of Theorem 1.1. The supplementary material is provided solely for independent checking, regression testing, and reproducibility.

1.7 Organization

Section 2 fixes notation and the asymptotic theorems used in the proof. Section 3 solves the common-profile LP. Section 4 proves the fractional cloning bound and the global fractional margin. Section 5 develops averaged and double-factor rounding. Section 6 proves polarization. Section 7 proves shifted-center gain completion. Section 8 handles the sparse-independent regime. Section 9 assembles the three regimes. Section 10 records corollaries, Section 11 discusses the proof and its limitations, and Section 12 describes future uses and open directions. Appendices A–D contain the algebraic details, divisibility correction, a pointer to the computational audits in the companion, and the self-contained proof of the list edge coloring theorem. Appendix E presents a curated selection of paper-level corollaries and reusable interfaces extracted from the formalization.

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2. Preliminaries

2.1 Split notation

Throughout,

$$G = (K \sqcup I, E)$$

is a split graph, where K is a clique of size p and $I = \{v_1, \dots, v_q\}$ is independent.

For $v_i \in I$, write

$$N_i = N(v_i) \cap K, \qquad d_i = |N_i|, \qquad S_i = K \setminus N_i, \qquad m_i = |S_i|.$$

Set

$$M = \sum_i m_i, \qquad S_2 = \sum_i m_i^2.$$

When q is close to $2p$, write

$$q = 2p - s. \tag{2.1}$$

2.2 Triangle packings

A fractional triangle packing is a nonnegative weight assignment to the triangles of G such that the total weight through every edge is at most one. Its maximum value is $\nu_3^*(G)$.

The dual is a fractional triangle cover: a nonnegative weight assignment to edges such that every triangle receives total edge weight at least one. LP duality gives equality between the two optimum values.

Throughout we write

$$\Phi(G) := |E(G)| - 2\nu_3(G)$$

for the integral triangle-packing defect; by (1.1), $\mathrm{cp}(G) \le \Phi(G)$.

2.3 Complete graph factorizations

Adopt the boundary conventions

$$\chi'(K_0) = \chi'(K_1) = 0.$$

For $t \ge 2$, the edge-chromatic number of a complete graph is

$$\chi'(K_t) = \begin{cases} t-1, & t \text{ even}, \\ t, & t \text{ odd}. \end{cases} \tag{2.2}$$

Thus $E(K_t)$ decomposes into $\chi'(K_t)$ matchings. The conventions for $t \le 1$ make the same language valid for empty edge sets.

2.4 Asymptotic theorems used in the proof

The written proof uses two asymptotic theorems from the literature. Their formal counterparts and verification status are recorded in the companion linked in Section 11.6. The list-edge-coloring case required here is proved self-contained in Appendix D. Thus Sections 5–7 and Proposition 10.5 use no asymptotic input, though they use standard facts such as the edge coloring of complete graphs.

Theorem 2.1 — Haxell–Rödl/Yuster [11,17]

For every fixed graph H,

$$\nu_H^*(G) - \nu_H(G) = o(|V(G)|^2)$$

uniformly over graphs G. We apply this with $H = K_3$.

Theorem 2.2 — List edge coloring (proved in Appendix D)

Let B be a simple bipartite graph with maximum degree $\Delta(B)$. If every edge e is assigned a list $L(e)$ with

$$|L(e)| \geq \Delta(B),$$

then B has a proper edge coloring choosing the color of each edge from its list.

This is the maximum-degree case of Galvin's theorem [10]. A self-contained proof (König coloring, stable pairings, and the kernel lemma) is given in Appendix D, so this is not an external dependency of the paper. The local refinement $|L(xy)| \geq \max\{d(x), d(y)\}$ of Borodin, Kostochka, and Woodall [4] is not needed: hypothesis (7.2) already bounds every list by the maximum degree of the gain graph.

Theorem 2.3 — Dense triangle decomposition

For every $\varepsilon > 0$, every sufficiently large triangle-divisible graph H with

$$\delta(H) \geq (0.9 + \varepsilon)|V(H)|$$

admits a triangle decomposition. This follows from Dross's fractional triangle decomposition theorem [7] (minimum degree $0.9v$ suffices fractionally) combined with the iterative-absorption theorem of Barber, Kühn, Lo, and Osthus [2], which converts the fractional threshold into an exact decomposition threshold for triangle-divisible graphs, up to ε and for sufficiently large order.

Remark 2.3′ — Necessity of a density hypothesis Triangle-divisibility alone does not imply triangle-decomposability. The 6-cycle has all degrees even and an edge count divisible by 3, but it has no triangle decomposition. A second obstruction is obtained from K_7 by deleting two vertex-disjoint triangles: the resulting graph is triangle-divisible and has 15 edges, but it is not triangle-decomposable. Thus a decomposition theorem for triangle-divisible graphs requires an additional hypothesis such as the density condition in Theorem 2.3. These examples motivate that hypothesis; they are not premises of Theorem 2.3 or Theorem 1.1.

2.5 Formalized auxiliary interfaces

The formal development records consequences beyond the main proof. Appendix E retains only those that sharpen the paper's principal constructions or are reusable outside the split-graph application: factorization packing bounds, selected corridor estimates, the exact common-profile value, the unified fractional margin, the complete-split sharpness package, finite LP duality, matching cleanup, and an r-uniform nibble interface. Implementation-level degree windows, typed-hypergraph encoding lemmas, monotonicity wrappers, and duplicate numerical forms remain available in the Lean sources but are not restated in the manuscript.

These interfaces are consequences or proof infrastructure, not additional premises of Theorem 1.1. Their release status is recorded in the companion document linked in Section 11.6.

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3. The Common-Profile Linear Program

For integers p, q, d, let $H(p,q,d)$ be the split graph with clique K, $|K| = p$, independent set I, $|I| = q$, and a fixed set $N \subseteq K$, $|N| = d$, such that every vertex of I has neighborhood N. Put $R = K \setminus N$ and $r = p - d$.

3.1 Symmetric cover variables

Averaging over permutations of N, R, and I, an optimal fractional triangle cover may be assumed constant on the four edge classes

$$E(N), \qquad E(N,I), \qquad E(N,R), \qquad E(R).$$

Let their weights be a, b, c, e, respectively. The triangle constraints are

$$a + 2b \geq 1, \qquad 3a \geq 1, \tag{3.1}$$

$$a + 2c \geq 1, \qquad 2c + e \geq 1, \qquad 3e \geq 1. \tag{3.2}$$

Constraints corresponding to empty triangle types may be omitted; the formula below remains valid for $p \geq 3$.

The objective is

$$\binom{d}{2} a + qdb + dr\, c + \binom{r}{2} e. \tag{3.3}$$

3.2 Exact solution

Theorem 3.1 — Common-profile formula

For $p \geq 3$ and $q \geq 1$,

$$\boxed{\nu_3^*(H(p,q,d)) = F(p,q,d)}, \tag{3.4}$$

where

$$\boxed{F(p,q,d) = \min\left\{ \frac{\binom{p}{2} + qd}{3}, \binom{d}{2} + \binom{r}{2}, \binom{d}{2} + \frac{dr + \binom{r}{2}}{3} \right\}}. \tag{3.5}$$

Proof

At an optimum,

$$b = \frac{1-a}{2}, \qquad c = \frac{1 - \min\{a, e\}}{2}, \qquad \frac{1}{3} \leq a, e \leq 1.$$

If $a \leq e$, decreasing e to a cannot increase the objective. It therefore suffices to minimize on

$$\frac{1}{3} \le e \le a \le 1.$$

The objective is affine on this triangle, so a minimum occurs at

$$(a, e) = \left(\frac{1}{3}, \frac{1}{3}\right), \qquad (1, 1), \qquad \left(1, \frac{1}{3}\right).$$

The corresponding values are precisely the three expressions in (3.5). Duality completes the proof. □

When $q = 0$, the graph $H(p, 0, d)$ is simply K_p, independently of d, and the same formula follows from $\nu_3^*(K_p) = \binom{p}{2}/3$.

3.3 Interpretation of the three covers

The minimum in (3.5) is easier to remember through the three vertices of the reduced two-variable program. Each vertex represents a distinct way to pay for the triangle constraints.

Cover pattern	Reduced vertex (a, e)	Value	Geometric description
Uniform	$(1/3, 1/3)$	$\left(\binom{p}{2} + qd\right)/3$	All relevant edge orbits are covered at the uniform fractional rate.
Separated	$(1, 1)$	$\binom{d}{2} + \binom{r}{2}$	The two clique blocks N and R are paid internally, with no weight on the crossing orbit.
Hot neighborhood	$(1, 1/3)$	$\binom{d}{2} + \left(dr + \binom{r}{2}\right)/3$	The common neighborhood N is paid in full, while the residual clique structure remains fractional.

The labels are descriptive rather than additional definitions. In later calculations we refer to the corresponding first, second, and third branches of F. This finite trichotomy is the source of the global fractional margin.

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4. Fractional Cloning and the Exact Fractional Margin

4.1 Fractional cloning

The fractional cloning operation (replication) used here is the cover-side analogue of the clone symmetrization used in Paper II. There, vertices or clone classes are copied pairwise to simplify a chordal graph. Here, one independent-set profile is copied q times in a single averaging step; the graph $H(p, q, d_i)$ records the resulting common profile.

Lemma 4.1 — Fractional cloning bound

For every split graph with $q \geq 1$,

$$\boxed{\nu_3^*(G) \geq \frac{1}{q} \sum_{i=1}^{q} F(p, q, d_i)}. \tag{4.1}$$

Proof

Let y be any fractional triangle cover of G. Put

$$A = \sum_{e \in E(K)} y_e$$

and

$$B_i = \sum_{x \in N_i} y_{v_i x}.$$

Replace v_i by q independent clones, all with neighborhood N_i, and give each clone the incident weights of v_i. Together with the original clique-edge weights, this is a fractional cover of $H(p, q, d_i)$ of weight $A + qB_i$. Hence

$$A + qB_i \geq F(p, q, d_i).$$

Summing over i gives

$$q\left(A + \sum_i B_i\right) \geq \sum_i F(p, q, d_i).$$

Minimizing over covers proves the lemma. □

4.2 The exact margin

Assume $0 < q \leq 2p$, set

$$\alpha = \frac{q}{p},$$

and define the packing threshold

$$T(G) = \frac{1}{2}\left(|E(G)| - \frac{(p+q)^2}{6}\right). \tag{4.2}$$

Define

$$\mu(\alpha) = \begin{cases} \alpha^2/12, & 0 \leq \alpha \leq 2/3, \\ (2-\alpha)^2/48, & 2/3 \leq \alpha \leq 2. \end{cases} \tag{4.3}$$

Theorem 4.2 — Unified fractional margin

$$\boxed{\nu_3^*(G) \geq T(G) + \mu(\alpha)p^2 - \frac{p}{4}}. \tag{4.4}$$

Proof

Let

$$C_\alpha = \frac{2 - 2\alpha - \alpha^2}{12}.$$

For each branch of $F(p, q, d)$, completion of squares gives

$$F(p, q, d) \geq \frac{qd}{2} + C_\alpha p^2 + \mu(\alpha)p^2 - \frac{p}{2}. \tag{4.5}$$

The three normalized residual minima are

$$\frac{\alpha^2}{12}, \qquad \frac{(2-\alpha)^2}{48},$$

and

$$\begin{cases} \alpha(8 - 5\alpha)/48, & \alpha \leq 4/3, \\ (2-\alpha)^2/12, & \alpha \geq 4/3. \end{cases}$$

The third is never below the minimum of the first two. Averaging (4.5) through Lemma 4.1 gives

$$\nu_3^*(G) \geq \frac{1}{2}\sum_i d_i + C_\alpha p^2 + \mu(\alpha)p^2 - \frac{p}{2}.$$

Since

$$T(G) = \frac{1}{2}\sum_i d_i + C_\alpha p^2 - \frac{p}{4},$$

we obtain (4.4). □

4.3 Bulk consequence

If

$$\varepsilon \leq \alpha \leq 2 - \varepsilon,$$

then $\mu(\alpha) \geq c_\varepsilon > 0$. Theorem 4.2 and Haxell–Rödl imply

$$\nu_3(G) \geq T(G)$$

for all sufficiently large graphs in this regime. Hence

$$|E(G)| - 2\nu_3(G) \leq \frac{n^2}{6}. \tag{4.6}$$

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5. Factorization Rounding Near $q = 2p$

Let

$$r_p = \chi'(K_p).$$

5.1 One-factor averaging

Lemma 5.1

If $q \geq r_p$, then

$$\boxed{\nu_3(G) \geq \frac{1}{q} \sum_i \binom{d_i}{2}}. \tag{5.1}$$

Proof

Factor K_p into r_p matchings. Assign the factors injectively and uniformly to vertices of I. In the factor assigned to v_i, retain only edges with both endpoints in N_i. The retained edges form valid edge-disjoint KKI triangles.

The expected number retained is

$$\frac{1}{q} \sum_i \sum_{j=1}^{r_p} |F_j \cap E(K[N_i])| = \frac{1}{q} \sum_i \binom{d_i}{2}.$$

Some assignment attains at least the expectation. □

Appendix E.1 records the assignment-level packing inequality before averaging, with the declaration correspondence in the formal companion.

Writing $q = 2p - s$, substitution gives

$$\Phi(G) \leq \frac{n^2}{6} + \frac{p}{2} - \frac{s^2}{6} + \frac{(s-1)M - S_2}{q}. \tag{5.2}$$

Using $S_2 \geq M^2/q$ and maximizing the resulting parabola yields

$$\boxed{\Phi(G) \leq \frac{n^2}{6} + \frac{p}{2} + \frac{s^2 - 6s + 3}{12}}. \tag{5.3}$$

Thus $s = O(\sqrt{p})$ is closed with linear error.

5.2 Double-factor rounding

For a clique edge $e = xy$, let

$$b_e = |\{i : \{x, y\} \not\subseteq N_i\}|.$$

Put

$$h = \min\{r_p, q - r_p\}, \qquad \delta = \frac{h}{r_p}.$$

Choose the set of h factors that receive two distinct independent vertices uniformly at random, give one independent vertex to every other factor, and assign the independent vertices to the resulting positions uniformly and injectively. All expectations below are taken over both random choices; some deterministic choice attains at least the expected gain.

Lemma 5.2 — Double-factor inequality

If $q \geq r_p$, then

$$\boxed{\Phi(G) \leq \frac{n^2}{6} + \frac{p}{2} - \frac{s^2}{6} + \frac{(s-1)M - S_2}{q} - \frac{2\delta V}{q(q-1)},} \tag{5.4}$$

where

$$V = \sum_{e \in E(K)} b_e(q - b_e). \tag{5.5}$$

Proof

If the factor containing e receives one vertex, e is lost with probability b_e/q. If it receives two vertices, it is lost only when both are bad, with probability

$$\frac{b_e(b_e - 1)}{q(q-1)}.$$

Thus the expected number U of lost clique edges is

$$U = \frac{1}{q}\sum_e b_e - \frac{\delta}{q(q-1)}\sum_e b_e(q - b_e).$$

Because every factor is a matching, each retained edge may be assigned to one of its admitting vertices without creating an edge collision. Substitution gives (5.4). □

The corresponding packing-form lower bound is stated as Corollary E.1.2.

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6. Quantitative Polarization

For every i, define

$$\mathcal{B}_i = \{e \in E(K) : e \cap S_i \neq \varnothing\}.$$

Then

$$V = \sum_{i,j} |\mathcal{B}_i \setminus \mathcal{B}_j|. \tag{6.1}$$

Let

$$m = \max_i |S_i|.$$

Lemma 6.1 — Polarization inequality

If $2p - 3m - 1 \geq 0$, then

$$\boxed{V \geq \frac{2p - 3m - 1}{4} \sum_{i,j} |S_i \triangle S_j|}. \tag{6.2}$$

Proof

Put

$$a_{ij} = |S_i \setminus S_j|.$$

The edges in $\mathcal{B}_i \setminus \mathcal{B}_j$ are precisely the edges of $K_p - S_j$ meeting $S_i \setminus S_j$. Therefore

$$|\mathcal{B}_i \setminus \mathcal{B}_j| = \frac{a_{ij}\big(2(p - |S_j|) - a_{ij} - 1\big)}{2}.$$

Since $|S_j| \leq m$ and $a_{ij} \leq m$,

$$|\mathcal{B}_i \setminus \mathcal{B}_j| \geq \frac{2p - 3m - 1}{2} |S_i \setminus S_j|.$$

Sum over ordered pairs and use

$$\sum_{i,j} |S_i \setminus S_j| = \frac{1}{2} \sum_{i,j} |S_i \triangle S_j|.$$

This proves (6.2). $\square$

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7. Shifted-Center Gain Completion

Fix $R \subseteq K$. Put

$$\rho = |R|, \qquad Q = K \setminus R, \qquad b = |Q|.$$

For each i, define

$$T_i = S_i \setminus R, \qquad t_i = |T_i|,$$

and

$$G_i = R \setminus S_i, \qquad g_i = |G_i|.$$

Set

$$A_R = \sum_i t_i, \qquad A_{2,R} = \sum_i t_i^2, \qquad B_R = \sum_i g_i.$$

Let

$$r_b = \chi'(K_b), \qquad u = q - r_b.$$

Assume

$$b \geq 2, \qquad q \geq r_b, \qquad b \geq \chi'(K_\rho), \tag{7.1}$$

and

$$b - t_i \geq \max\{\rho, u\} \qquad \text{for all } i. \tag{7.2}$$

Define

$$\theta_R = \frac{\max\{\rho - 1, 0\}}{b}$$

and

$$\kappa_R = 1 - 2(1 - \theta_R)\frac{u}{q}. \tag{7.3}$$

7.1 Packing inside Q

Reserve a set $U \subseteq I$ of u vertices; note that $|I \setminus U| = q - u = r_b$ exactly. Assign the remaining r_b vertices bijectively to a factorization of $K[Q]$.

For v_i, the number of unavailable edges of $K[Q]$ is

$$\beta_i = \binom{b}{2} - \binom{b - t_i}{2}. \tag{7.4}$$

For fixed U, averaging over assignments gives at least

$$\binom{b}{2} - \frac{1}{r_b} \sum_{i \notin U} \beta_i \tag{7.5}$$

edge-disjoint QQI triangles.

7.2 The gain graph

Construct a bipartite graph with parts U and R, joining v_i to r when $r \in G_i$. Assign the list

$$L(v_i r) = N_i \cap Q.$$

By (7.2), every list satisfies

$$|L(v_i r)| = b - t_i \geq \max\{\rho, u\} \geq \Delta,$$

where Δ is the maximum degree of the gain graph, since $d(v_i) = g_i \leq \rho$ and $d(r) \leq |U| = u$. Theorem 2.2, proved self-contained in Appendix D, gives a proper list edge coloring. If $v_i r$ receives color $z \in Q$, take the triangle $v_i r z$. This gives

$$B_U = \sum_{i \in U} g_i$$

edge-disjoint IRQ triangles.

7.3 Completion inside R

For each $z \in Q$, let $U_z \subseteq R$ be the vertices r for which rz was used by an IRQ triangle.

If $\rho \leq 1$, the graph $K[R]$ has no edges, so this completion contributes no RRQ triangles. Suppose therefore that $\rho \geq 2$. Factor $K[R]$, inject its factors into the colors $z \in Q$, and delete from the factor assigned to z every edge incident with U_z. Averaging over the injections loses at most

$$\frac{\rho - 1}{b} \sum_z |U_z| = \theta_R B_U$$

edges. Hence, in all cases, we obtain at least

$$\binom{\rho}{2} - \theta_R B_U$$

additional RRQ triangles.

The three families QQI, IRQ, and RRQ are edge-disjoint. In particular, the forbidden-color deletion prevents an IRQ triangle and an RRQ triangle from sharing an edge rz.

7.4 The centered inequality

Choose U, $|U| = u$, maximizing

$$\sum_{i \in U} \left(\frac{2\beta_i}{r_b} + 2(1 - \theta_R) g_i \right).$$

The best u terms have sum at least u/q times the total. Using

$$2 \sum_i \beta_i = (2b - 1) A_R - A_{2,R},$$

we obtain the principal local lemma. The subset-host packing form is recorded as Corollary E.1.3.

Lemma 7.1 — Reserved-gain shifted-center inequality

Under (7.1)–(7.2),

$$\boxed{\begin{aligned} \Phi(G) \le \frac{n^2}{6} + \frac{p}{2} - \frac{s^2}{6} + s\rho - 2\rho^2 \\ + \kappa_R B_R + \frac{(s - 2\rho - 1) A_R - A_{2,R}}{q}. \end{aligned}} \tag{7.6}$$

8. The Sparse-Independent Regime

Assume $q = o(p)$. We prove the required bound directly and integrally.

8.1 Minimal-counterexample degree bound

In the global contradiction argument, a minimal counterexample with penalty kn satisfies, for every $v \in I$,

$$d(v) > \frac{2n - 1}{6} + k. \tag{8.1}$$

Since $q = o(p)$, this implies eventually

$$d(v) \ge 2q + 2. \tag{8.2}$$

8.2 Successive matchings

Order $I = \{v_1, \dots, v_q\}$. Choose successively edge-disjoint matchings

$$M_i \subseteq K[N_i]$$

of size $\lfloor d_i/2 \rfloor$.

Before choosing M_i, each vertex has lost at most $i-1$ incident edges, so the available graph on N_i has minimum degree at least

$$d_i - i \geq d_i/2.$$

Dirac's theorem [6] supplies a Hamilton cycle and hence a matching of the required size.

Let

$$F = \bigcup_i M_i.$$

Then

$$|F| \geq \frac{1}{2}\sum_i d_i - \frac{q}{2}, \qquad \Delta(F) \leq q. \tag{8.3}$$

The edges of M_i yield $|M_i|$ edge-disjoint KKI triangles with apex v_i.

8.3 Divisibility correction

The residual clique graph

$$R_0 = K_p - F$$

satisfies

$$\delta(R_0) \geq p - 1 - q = (1 - o(1))p. \tag{8.4}$$

In particular, for all sufficiently large members of the sequence, $\delta(R_0) > p/2$. Dirac's theorem [6] therefore gives a Hamilton cycle, and hence a Hamilton path

$$P = x_1 x_2 \cdots x_p$$

contained in R_0. Let O be the set of odd-degree vertices of R_0. By the handshaking lemma, $|O|$ is even. Define the path subgraph $J \subseteq P$ by

$$x_j x_{j+1} \in E(J) \quad \Longleftrightarrow \quad |O \cap \{x_1, \dots, x_j\}| \text{ is odd.} \tag{8.5}$$

For an internal vertex x_j, the parity of its degree in J is the change in prefix parity between positions $j-1$ and j, and is therefore 1 precisely when $x_j \in O$. The same conclusion holds at the two endpoints because $|O|$ is even. Consequently,

$$\mathrm{Odd}(J) = O, \qquad |E(J)| \le p-1, \qquad \Delta(J) \le 2. \tag{8.6}$$

Thus

$$R_1 := R_0 - J$$

has all degrees even, and

$$\delta(R_1) \ge p - 1 - q - 2. \tag{8.7}$$

Since $q = o(p)$, eventually $\delta(R_1) > 3p/4$. By Turán's theorem, R_1 contains a copy of K_5. Choose a C_4 and a C_5 inside this fixed K_5. According as

$$|E(R_1)| \equiv 0, 1, 2 \pmod 3,$$

remove respectively nothing, the C_4, or the C_5. Let C denote the removed cycle, allowing $C = \varnothing$, and put

$$H := R_1 - E(C).$$

Every vertex loses either zero or two incident edges when a cycle is removed, so all degrees of H remain even. Moreover, $|E(C)| \equiv |E(R_1)| \pmod 3$, and hence

$$|E(H)| \equiv 0 \pmod 3.$$

Therefore H is triangle-divisible. The total loss of degree in passing from R_0 to H is at most four, so

$$\delta(H) \ge p - 1 - q - 4. \tag{8.8}$$

To apply Theorem 2.3 with a fixed parameter, take for example $\varepsilon_0 = 1/100$. Since $q = o(p)$, eventually $q \le p/20$, and then, for sufficiently large p,

$$\delta(H) \ge p - 1 - q - 4 \ge 0.91p = (0.9 + \varepsilon_0)|V(H)|. \tag{8.9}$$

Theorem 2.3 now gives an exact triangle decomposition of H. Notice that the decomposition theorem is applied on the original p-vertex set: no vertices are deleted during the correction. The entire correction removes at most $p + 4$ edges.

8.4 Packing size

The combined packing has size

$$\begin{aligned}\nu_3(G) &\geq |F| + \frac{\binom{p}{2} - |F| - O(p)}{3} \\ &= \frac{1}{3}\binom{p}{2} + \frac{2}{3}|F| - O(p) \\ &\geq \frac{1}{3}\binom{p}{2} + \frac{1}{3}\sum_i d_i - O(p+q).\end{aligned} \tag{8.10}$$

Consequently,

$$\begin{aligned}\Phi(G) &\leq \frac{1}{3}\binom{p}{2} + \frac{1}{3}\sum_i d_i + O(p+q) \\ &\leq \frac{1}{3}\binom{p}{2} + \frac{pq}{3} + O(p+q) \\ &= \frac{(p+q)^2}{6} - \frac{p+q^2}{6} + O(p+q).\end{aligned} \tag{8.11}$$

This is at most $n^2/6 + O(n)$. Appendix E.6 records the formalized matching cleanup and bounded divisibility-correction interfaces used in this regime.

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9. Proof of the Main Theorem

We now prove Theorem 1.1 by contradiction.

Suppose no absolute constant works. For every positive integer k, choose a split graph G_k, minimal in its number n_k of vertices, such that

$$\Phi(G_k) > \frac{n_k^2}{6} + kn_k. \tag{9.1}$$

Necessarily $n_k \to \infty$. For every $v \in I(G_k)$, minimality gives

$$\Phi(G_k) \leq \Phi(G_k - v) + d(v),$$

and therefore

$$\boxed{d(v) > \frac{2n_k - 1}{6} + k}. \tag{9.2}$$

First, if $q_k \geq 2p_k - 1$, Lemma 5.1 gives

$$\Phi(G_k) \leq \frac{n_k^2}{6} + \frac{p_k}{2},$$

contradicting (9.1) for large k. We may therefore assume $q_k < 2p_k - 1$ and pass to a subsequence according to

$$\alpha_k = \frac{q_k}{p_k} \in [0, 2).$$

9.1 Bulk regime

Suppose that for some $\varepsilon > 0$,

$$\varepsilon \le \alpha_k \le 2 - \varepsilon.$$

Theorem 4.2 gives

$$\nu_3^*(G_k) \ge T(G_k) + c_\varepsilon p_k^2 - O(p_k).$$

Haxell–Rödl gives

$$\nu_3(G_k) \ge \nu_3^*(G_k) - o(n_k^2).$$

For sufficiently large k, the quadratic margin dominates the integrality loss, so $\nu_3(G_k) \ge T(G_k)$, contrary to (9.1).

9.2 The endpoint $\alpha_k \to 0$

This is closed by Section 8, which gives an absolute C_0 such that

$$\Phi(G_k) \le \frac{n_k^2}{6} + C_0 n_k.$$

For $k > C_0$, this contradicts (9.1).

9.3 The endpoint $\alpha_k \to 2$

Write

$$q = 2p - s, \qquad s = o(p).$$

If $s = O(\sqrt{p})$, inequality (5.3) gives the required linear-error bound. Assume henceforth

$$\sqrt{p} \ll s = o(p). \tag{9.3}$$

For all sufficiently large members of the sequence,

$$p \ge 2304, \qquad 6\sqrt{p} \le s \le \frac{p}{8}, \qquad k \ge 1. \tag{9.4}$$

By (9.2),

$$m_i = |S_i| < \frac{s}{3} + \frac{1}{6} - k \le \frac{s}{3} - \frac{5}{6}.$$

Since every m_i is an integer, with $m = \max_i m_i$,

$$\boxed{3m \le s - 3}. \tag{9.5}$$

For $x \in K$, put

$$a_x = |\{i : x \in S_i\}|$$

and

$$D = \sum_{x \in K} a_x(q - a_x). \tag{9.6}$$

Then

$$\sum_{i,j} |S_i \triangle S_j| = 2D. \tag{9.7}$$

High dispersion

Suppose

$$D \geq \frac{qs^2}{12}. \tag{9.8}$$

By (9.5),

$$2p - 3m - 1 \geq 2p - s + 2 = q + 2.$$

Lemma 6.1 and (9.7) therefore give

$$V \geq \frac{q+2}{2} D \geq \frac{q}{2} D. \tag{9.9}$$

The double-factor coefficient in Lemma 5.2 satisfies

$$\boxed{\delta \geq \frac{7}{8}}. \tag{9.10}$$

Indeed, if p is odd then $r_p = p$ and

$$\delta = \frac{p - s}{p} \geq \frac{7}{8},$$

while if p is even then $r_p = p - 1$ and

$$\delta = \frac{p + 1 - s}{p - 1} \geq \frac{7}{8}.$$

Moreover, $S_2 \geq M^2/q$, and (9.5) implies $M/q \leq m \leq (s-3)/3$. Since the resulting parabola is increasing on this interval,

$$\frac{(s-1)M - S_2}{q} \leq \frac{2s(s-3)}{9}. \tag{9.11}$$

Substituting (9.8)–(9.11) into Lemma 5.2 yields

$$\Phi(G) - \frac{n^2}{6} \le \frac{p}{2} - \frac{5s^2}{288} - \frac{2s}{3}. \tag{9.12}$$

Since $s^2 \ge 36p$,

$$\Phi(G) - \frac{n^2}{6} \le -\frac{p}{8} - \frac{2s}{3} < 0,$$

a contradiction.

Low dispersion

Suppose

$$D < \frac{qs^2}{12}. \tag{9.13}$$

By (9.7), some j satisfies

$$\sum_i |S_i \triangle S_j| < \frac{s^2}{6}. \tag{9.14}$$

Set

$$R = S_j, \qquad \rho = |R|.$$

Then

$$A_R + B_R < \frac{s^2}{6}, \qquad \rho \le m \le \frac{s-3}{3}. \tag{9.15}$$

We verify the hypotheses of Lemma 7.1 explicitly. Since $s \le p/8$,

$$q = 2p - s \ge \frac{15p}{8}, \qquad b = p - \rho \ge p - \frac{s}{3}.$$

Thus $b \ge 2$, $q \ge r_b$, and $b \ge \chi'(K_\rho)$. Also $r_b \ge b - 1$, so

$$u = q - r_b \le p - s + \rho + 1.$$

For every i,

$$2\rho + t_i + 1 \le 3m + 1 \le s - 2.$$

Consequently,

$$b - t_i \ge \max\{\rho, u\},$$

and Lemma 7.1 applies.

If $\rho = 0$, then $B_R = 0$. If $\rho \geq 1$, then

$$\theta_R \leq \frac{\rho}{p-\rho} \leq \frac{8s}{23p}.$$

Using $u = q - r_b$, $r_b \leq b$, and $q \geq 15p/8$, we obtain

$$\kappa_R = 1 - 2(1-\theta_R)\frac{u}{q} \leq \frac{s}{q} + 2\theta_R \leq \frac{5s}{4p}. \tag{9.16}$$

Likewise,

$$\frac{(s-2\rho-1)_+}{q} \leq \frac{5s}{4p}. \tag{9.17}$$

By (9.15), the total positive deviation in (7.6) is therefore at most

$$\frac{5s}{4p}(A_R + B_R) < \frac{5s^3}{24p} \leq \frac{5s^2}{192}. \tag{9.18}$$

Meanwhile,

$$\frac{s^2}{6} - s\rho + 2\rho^2 = 2\left(\rho - \frac{s}{4}\right)^2 + \frac{s^2}{24} \geq \frac{s^2}{24}. \tag{9.19}$$

Lemma 7.1 now gives

$$\Phi(G) - \frac{n^2}{6} \leq \frac{p}{2} - \frac{s^2}{64}. \tag{9.20}$$

Since $s^2 \geq 36p$, the right-hand side is at most $-p/16$, again a contradiction.

All possible subsequences are impossible. Theorem 1.1 follows. □

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10. Corollaries

10.1 Clique partitions

Corollary 1.2 follows immediately from

$$\mathrm{cp}(G) \leq |E(G)| - 2\nu_3(G).$$

10.2 Sharpness of the leading term

The family

$$K_p \vee \overline{K}_{2p}$$

has $n = 3p$. A factorization of K_p assigned to independent vertices packs every clique edge into a KKI triangle. The resulting triangle-packing expression is

$$|E| - 2\nu_3 = \frac{n^2}{6} + \frac{n}{6}. \tag{10.1}$$

The Erdős–Ordman–Zalcstein clique-partition construction on this family shows that the coefficient $1/6$ of the quadratic term cannot be improved. The exact identity also gives a separate statement about the linear term: if

$$\mathrm{cp}(G) \le \frac{n^2}{6} + Cn$$

holds uniformly over all split graphs, then necessarily $C \ge 1/6$. These two roles of $1/6$ are kept distinct below; the latter lower bound does not assert that $C = 1/6$ is sufficient for every split graph.

Corollary 10.2a — Exact complete-split benchmark

For every $p \ge 1$,

$$\mathrm{cp}\big(K_p \vee \overline{K}_{2p}\big) = 2p^2 - \binom{p}{2} = \frac{(3p)^2}{6} + \frac{3p}{6}. \tag{10.1a}$$

Proof

The factorization packing above uses every clique edge in a KKI triangle. Retaining those triangles and placing each remaining edge in a K_2 gives a clique partition with $2p^2 - \binom{p}{2}$ parts.

For the reverse inequality, assign weight $+1$ to every cross edge and weight -1 to every clique edge. The total edge weight is $2p^2 - \binom{p}{2}$. A clique contains at most one independent vertex; if it contains t clique vertices and one independent vertex, its edge weight is $t - \binom{t}{2} \le 1$, while a clique contained in K_p has nonpositive weight. Every part of a clique partition therefore contributes at most one to the total weight. Hence every clique partition has at least $2p^2 - \binom{p}{2}$ parts.

Appendix E.5 retains only the exact normal form and the logically distinct forced-linear-coefficient statement.

Corollary 10.2b — Sharp quadratic term and forced linear coefficient

There is an absolute constant C such that

$$\mathrm{cp}(G) \le \frac{n^2}{6} + Cn$$

for every split graph G. Every constant C satisfying this uniform estimate obeys $C \geq 1/6$, and for every $p \geq 1$ the complete-split graph $K_p \vee \overline{K}_{2p}$, on $n = 3p$ vertices, satisfies

$$\mathrm{cp}\big(K_p \vee \overline{K}_{2p}\big) = \frac{n^2}{6} + \frac{n}{6}.$$

In particular, $n^2/6$ is the sharp quadratic term. The statement gives a necessary lower bound on the admissible linear coefficient; it does not identify the least uniform linear coefficient. Determining an explicit, and ultimately the least, admissible coefficient is part of Problem 12.2.

Proof

Existence follows from Corollary 1.2, and the exact identity is Corollary 10.2a. If the displayed uniform estimate holds for a constant C, applying it to $K_1 \vee \overline{K}_2$ and using the exact identity gives $2 \leq 3/2 + 3C$, hence $C \geq 1/6$. The complete-split family also witnesses sharpness of the quadratic coefficient as $p \to \infty$.

10.3 Retrospective on the corridor decomposition

The localization in Section 1.3 is an internal step of the contradiction proof, not a surviving obstruction after Theorem 1.1. The bulk margin excludes ratios bounded away from 0 and 2, the sparse construction excludes $|I|/|K| \to 0$, and averaged factorization excludes the short corridor near $|I| = 2|K|$. These reductions explain why the load-bearing part of the proof is the mesoscopic corridor

$$|I| = 2|K| - s, \qquad \sqrt{|K|} \ll s = o(|K|),$$

which is eliminated by the dispersion dichotomy.

10.4 Exact common-profile fractional value

Theorem 3.1 is independently useful: it gives the exact fractional triangle packing for every split graph in which all independent vertices have one common neighborhood, including the regime with too few independent vertices to color all neighborhood edges integrally. We state it as a stand-alone corollary.

Corollary 10.4a — Exact common-neighborhood triangle packing

For all integers $p \geq 3$, $q \geq 1$ and $0 \leq d \leq p$, the graph $H(p, q, d)$ (clique of order p; q independent vertices, each with the same neighborhood of size d) satisfies

$$\nu_3^*(H(p,q,d)) = F(p,q,d) = \min\left\{ \tfrac{\binom{p}{2}+qd}{3},\ \binom{d}{2} + \binom{p-d}{2},\ \binom{d}{2} + \tfrac{d(p-d)+\binom{p-d}{2}}{3} \right\}.$$

The same identity holds for $q = 0$, where $H(p, 0, d) = K_p$ and $\nu_3^*(K_p) = \binom{p}{2}/3$. This is an exact closed form for the fractional triangle-packing number of an entire one-parameter family, valid for every d, including the small-q regime where no integral factorization covers the neighborhood. Appendix E.3 records the formalized branch comparisons and explicit phase regions.

Corollary 10.4b — Threshold graphs

Since $K_p \vee \overline{K}_{2p}$ is a threshold graph and every threshold graph is split, Theorem 1.1 gives, in particular, the linear-error bound $\mathrm{cp}(G) \leq n^2/6 + Cn$ for all threshold graphs, with the same sharp leading constant witnessed inside that subclass.

10.5 Localized effectivity near $q = 2p$

Packing-form corollaries

The following corollaries restate results from Sections 5 and 7 directly in packing language. All three are unconditional: none uses Theorem 2.1 or Theorem 2.3.

Corollary 5.1a (max-over-assignments). For a factorization $E(K_p) = F_1 \sqcup \cdots \sqcup F_{r_p}$ and $q \geq r_p$,

$$\nu_3(G) \;\geq\; \max_{\sigma:[r_p]\hookrightarrow I} \sum_{j=1}^{r_p} \big|F_j \cap E\big(K[N_{\sigma(j)}]\big)\big|.$$

Lemma 5.1 is the averaged consequence; the max form retains information lost in the average and is what an exact inequality needs. Because the assignment $\sigma : [r_p] \hookrightarrow I$ is injective, this bound uses at most r_p hosts. It is therefore most naturally used when q is comparable to r_p; when q is substantially larger, cyclic reuse of factors may exploit additional hosts, as in the corridor constructions later in the paper.

Corollary 5.2a (packing-native double factorization). With $b_e = \#\{i : e \not\subseteq N_i\}$, $V = \sum_{e\in E(K)} b_e(q - b_e)$, $h = \min\{r_p, q - r_p\}$ and $\delta = h/r_p$,

$$\nu_3(G) \;\geq\; \frac{1}{q}\sum_i \binom{d_i}{2} + \frac{\delta V}{q(q-1)}.$$

This exhibits the polarization gain of Lemma 5.2 as an integral packing gain, not merely an improvement of an auxiliary function. When $q = r_p$, one has $h = 0$, so the gain term vanishes and the statement reduces to the averaged bound of Lemma 5.1.

Proposition 7.4 (robust shifted-center packing bound). Let $R \subseteq K$, $Q = K \setminus R$, $\rho = |R|$, $b = |Q|$; for i write $g_i = |N_i \cap R|$, $h_i = |N_i \cap Q|$, $t_i = b - h_i$. Choose $J \subseteq I$ with $q_J = |J| \geq r_b = \chi'(K_b)$, $b \geq \chi'(K_\rho)$, and $h_i \geq \max\{\rho,\, q_J - r_b\}$ for every $i \in J$. Then

$$\nu_3(G) \;\geq\; \binom{b}{2} + \binom{\rho}{2} - \frac{(2b-1)A_J - A_{2J}}{2q_J} + (1-\theta_R)\frac{(q_J - r_b)B_J}{q_J},$$

with $A_J = \sum_{i\in J} t_i$, $A_{2J} = \sum_{i\in J} t_i^2$, $B_J = \sum_{i\in J} g_i$. The construction of Section 7 is applied to the retained hosts J only, so it survives deleting deficient hosts; the cover cost of the excluded hosts is not part of this packing bound and belongs to any covering application built on it.

Scope. These are exact packing statements. Theorem 4.2 is not used in an exact inequality: its role is to create quadratic slack in the bulk, not to provide an exact bound. Likewise, a $\tau_3 \leq 2\nu_3$ statement of Tuza type for split graphs is not a corollary of this paper and is treated separately.

Proposition 10.5 — Explicit corridor bounds

Let $q = 2p - s$ and $n = p + q$.

1. If

$$p \geq 37, \qquad 0 \leq s \leq 6\sqrt{p},$$

then

$$\boxed{\Phi(G) \leq \frac{n^2}{6} + \frac{3n}{2}}. \tag{10.2}$$

2. If

$$p \geq 2304, \qquad 6\sqrt{p} \leq s \leq \frac{p}{8},$$

and every $v \in I$ satisfies

$$d(v) > \frac{2n-1}{6} + 1,$$

then

$$\boxed{\Phi(G) \leq \frac{n^2}{6}}. \tag{10.3}$$

Consequently, for every $C \geq 2$, a minimum-order counterexample to

$$\Phi(G) \leq \frac{n^2}{6} + Cn$$

cannot satisfy

$$p \geq 2304, \qquad 0 \leq s \leq \frac{p}{8}.$$

Proof

For the short corridor, inequality (5.3) together with $s^2 \leq 36p$ gives $\Phi(G) - \frac{n^2}{6} \leq \frac{7p}{2} - \frac{s}{2} + \frac{1}{4}$. Since $p \geq 37$ and $s^2 \leq 36p$, we have $s \leq p-1$: otherwise $s \geq p$ would force $p^2 \leq s^2 \leq 36p$, hence $p \leq 36$. Therefore $4s + 1 \leq 4p$, which is equivalent to $\frac{7p}{2} - \frac{s}{2} + \frac{1}{4} \leq \frac{3}{2}(3p - s) = \frac{3}{2}n$, so

$$\Phi(G) \leq \frac{n^2}{6} + \frac{3n}{2}.$$

The mesoscopic assertion is exactly the quantitative high/low-dispersion calculation in Section 9.3. The final consequence follows from the degree inequality obtained by deleting an independent vertex from a minimum-order counterexample. □

Appendix E.2 records the short-corridor constant and the packing-to-partition bridge. The mid- and high-corridor conclusions are already stated above and are not duplicated in the appendix.

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11. Discussion

11.1 From fractional asymptotics to a linear-error integral theorem

The central difficulty is that the general theorem

$$\nu_3^*(G) - \nu_3(G) = o(n^2)$$

has no linear rate. The present proof avoids demanding one uniform rounding mechanism.

- In the bulk, the common-profile LP produces quadratic slack.
- Near $q = 2p$, explicit factorization structures replace general rounding.
- Near $q = 0$, the residual clique becomes dense enough for exact decomposition.

This regime-dependent strategy is weaker than a universal linear integrality-gap theorem but sufficient for the extremal clique-partition problem. The packing-versus-covering theme for triangles is classical; Tuza's conjecture and its fractional forms, for which Krivelevich [14] proved two fractional variants and an integral special case, provide the closest general context, though they are not used in the present proof.

11.2 Why residual maximum degree is not the right invariant

Early exploratory work attempted to prove that an optimal packing leaves a clique residual of bounded maximum degree. Numerical behavior depended strongly on the packing algorithm, and no such property is needed in the final proof.

The effective invariants are instead:

- the common-profile fractional margin;
- the variance term S_2;
- the polarization energy V;
- the centered deviations A_R, B_R;
- divisibility of the dense residual.

11.3 Localized effectivity and remaining non-effectivity

Proposition 10.5 shows that the entire near-extremal corridor treated by factorization, polarization, and shifted-center completion is quantitatively effective. One may take

$$C_{\mathrm{corr}} = \frac{3}{2}, \qquad p_0 = 2304, \qquad s_0(p) = 6\sqrt{p}, \qquad \eta_0 = \frac{1}{8}.$$

The global constant C remains non-effective for two independent reasons.

1. The Haxell–Rödl theorem is used through an asymptotic $o(n^2)$ statement in the bulk.
2. Dense triangle decomposition theorems are invoked with unspecified sufficiently-large thresholds in the sparse-independent regime.

Thus no hidden non-effectivity remains in Sections 5–7; it comes only from the two external asymptotic inputs used elsewhere in the proof. After Appendix D, Sections 5–7, the high/low-dispersion argument of Section 9.3, and Proposition 10.5 are self-contained and effective. Proposition 10.5 uses no instance of Theorem 2.1 or Theorem 2.3; its only external background is the standard edge coloring of complete graphs together with the list edge coloring theorem proved in Appendix D. Dirac's and Turán's theorems belong to the separate sparse-independent argument of Section 8.

11.4 Relationship with the fractional extremal theorem

The complete-split extremal theorem determines the global fractional maximum of

$$|E(G)| - 2\nu_3^*(G)$$

over both split and chordal graphs. The common-profile LP in the present paper serves a different purpose: it is a profile-sensitive local lower bound for the fractionally cloned graph, and its quantitative slack absorbs integrality loss away from the extremal corridor.

Thus the global fractional extremal calculation and the local common-profile calculation are compatible but logically distinct. The present proof is self-contained apart from the external theorems stated in Section 2.

The local calculation is the same three-orbit pure-profile program isolated in Paper I [15], expressed after adding the pure-profile Phase-I contribution. Under the identification $d = s$ and $p - d = o$, the three branches of $F(p, q, d)$ are $qd/2 + U$, $qd/2 + D$, and $qd/2 + H$, while $C_\alpha p^2 = R(p, q)$ for $\alpha = q/p$. Here the calculation is used to produce profile-sensitive quadratic slack for the integral argument.

Remark (cloning terminology). Lemma 4.1 and the clone-symmetrization step of Paper II use the same underlying replication operation in different roles. Paper II uses pairwise cloning to move a chordal graph toward a complete-split terminal graph. Lemma 4.1 uses q-fold profile replication on the cover side to compare an arbitrary profile with a common-profile graph. The established term *fractional cloning* is retained for the present inequality. Paper IV uses the term *clone step* for a different operation, the vertex-copy substitution that replaces one neighborhood rather than replicating a profile.

11.5 Relationship with the chordal linear-error problem

The complete-split reduction settles the asymptotically sharp chordal bound

$$\mathrm{cp}(G) \le \left(\frac{1}{6} + o(1)\right) n^2.$$

The stronger estimate

$$\mathrm{cp}(G) \le \frac{n^2}{6} + O(n)$$

is established for general chordal graphs in the subsequent Paper IV [IV]. The following three directions record the structural questions motivated by this split-graph proof; they are not a claim that the chordal bound remains unresolved.

1. **Stability of the fractional extremal theorem.** Upgrade the exact chordal fractional theorem of Paper II to a stability statement: every chordal graph whose value is within δn^2 of the fractional maximum is within εn^2 edge edits of a maximizing complete split graph. Away from near-extremality, the quadratic fractional slack absorbs the general integrality loss exactly as in Section 4.
2. **Robust corridor mechanisms.** Extend the factorization, polarization, and shifted-center arguments of Sections 5–7 from split graphs to εn^2-perturbations of complete split graphs, so that they apply to the near-extremal graphs produced by stability.
3. **Clique-tree assembly.** Patch the local estimates across the clique tree by an edge-ownership ledger that preserves separator capacities, in the spirit of shifted-center completion, so that no edge is charged twice.

An implementation of the clique-tree approach in item 3 would have to preserve edge ownership and separator capacities.

These directions describe possible uses of the present mechanisms, not the proof architecture of Paper IV. No result here depends on a later paper or on a particular clique-separator construction.

11.6 Formalization and reproducibility

The accompanying formalization and reproducibility document identifies the unchanged Lean/Mathlib freeze, its theorem interfaces and the recorded build and axiom evidence. This

editorial edition changes neither the mathematical results nor the frozen sources; no Lean/Lake rebuild was performed. Computational checks are supplementary evidence, not premises of the proofs.

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12. Potential Uses and Future Directions

12.1 Universal linear integrality gap

Problem 12.1

Is there an absolute constant C such that

$$\nu_3^*(G) - \nu_3(G) \le C|V(G)|$$

for every split graph G?

The present paper supplies supporting structures near the extremal corridor but does not resolve the bulk regime without quadratic slack.

12.2 Effective global constants

Problem 12.2

Find an explicit admissible global constant C in Theorem 1.1, and determine or bound the least such constant.

The near-extremal corridor is already effective by Proposition 10.5. A fully quantitative global theorem therefore requires only explicit rates in fractional-to-integral packing, or a structured replacement for Haxell–Rödl in the bulk, together with effective dense-decomposition thresholds in the sparse-independent regime.

12.3 Algorithmic packing

The proof contains polynomially implementable pieces:

- complete-graph factorizations;
- weighted selection of reserved vertices;
- bipartite list edge coloring;
- matching extraction;
- dense decomposition algorithms implicit in iterative absorption.

It is natural to seek a polynomial-time algorithm that outputs a clique partition of size

$$\frac{n^2}{6} + O(n)$$

with an explicit constant. The factorization, list-coloring, and matching steps of Sections 5–7 are individually polynomial; assembling them into a single approximation algorithm with a proven guarantee is left to future work.

Algorithmic remark 12.3 — Effective corridor construction

In the near-extremal corridor covered by Proposition 10.5 (explicit p_0, s_0), the proof is constructive at the level of its mathematical ingredients: it uses one- and double-factorizations of K_p, reserved-vertex selection, list edge coloring, and matching extraction. With explicit representations of these objects and standard implementations of the underlying routines, the construction is expected to run in polynomial time and to output a clique partition of size $n^2/6 + 3n/2$. The manuscript does not yet fix an input model, data structures, or a complete implementation-level complexity proof, so no standalone polynomial-time theorem is claimed here. Outside the corridor, the guarantee also relies on the non-constructive bulk input (Theorem 2.1).

12.4 Stability and extremal classification

Problem 12.4

Characterize split graphs satisfying

$$\mathrm{cp}(G) \geq \frac{n^2}{6} - o(n^2).$$

The proof suggests that near-extremal graphs should have $|I| = (2+o(1))|K|$ and low-dispersion absence profiles close to a common center.

Two pieces of evidence support a Simonovits-type stability statement. First, the supplied audit report records an exhaustive enumeration at $n = 9$ (all split graphs with $|K| = 3$). In that computation, graphs within an additive constant of the maximum lie within bounded edit distance of the complete-split family, while genuinely dispersed profiles fall a linear amount below the maximum. Second, the fractional margin of Theorem 4.2 already yields a level-set bound: away from near-extremality the quadratic slack $\mu(\alpha)p^2$ is strictly positive, so any near-maximizer must have α near the extremal ratio and small profile dispersion. A full stability theorem would still require quantifying the gain of a single symmetrization step and controlling the directions along which Φ^* is flat; we leave it as an open problem, most naturally phrased in the cut (graphon) metric to absorb the choice of nearest complete-split graph.

12.5 Higher-clique linear-error packing

The complete-split reduction determines the fractional extremal problem and the first-order integral asymptotics for fixed K_r-and-edge partitions. The natural higher-clique analogue of the present paper is the following linear-error question.

For fixed $r \geq 3$, does every split graph satisfy

$$\pi_r(G) \leq \frac{r-1}{4r} n^2 + O_r(n),$$

where $\pi_r(G)$ is the minimum number of parts in an edge partition into copies of K_r and single edges?

The common-profile and cloning ideas suggest one possible approach, but the relevant local orbit programs and integral rounding mechanisms become substantially more complicated. Design and decomposition methods such as those developed by Keevash [12] and the regular-slice framework of Allen, Böttcher, Cooley, and Mycroft [1] may be useful tools; they are not ingredients of the present proof.

12.6 Chordal and clique-tree applications

Shifted-center completion was designed to separate owned clique edges from protected interface edges. A parametric version may be useful in further chordal constructions, where separators are shared between neighboring maximal cliques.

12.7 Reusable proof principle

The paper illustrates a broader strategy:

> Use exact symmetric LPs to create slack away from extremality, then reserve explicit combinatorial rounding only for the corridor where the slack vanishes.

This may apply to other structured packing-covering problems in which a general regularity theorem is too weak at second order.

Appendix A. Algebra of the Fractional Margin

Let $x = d/p$ and $\alpha = q/p$. Ignoring the exact linear terms temporarily, the three normalized branches of F are

$$f_1(x) = \frac{1}{6} + \frac{\alpha x}{3},$$

$$f_2(x) = x^2 - x + \frac{1}{2},$$

and

$$f_3(x) = \frac{1}{6} + \frac{x^2}{3}.$$

Subtracting the affine target contribution gives minima

$$\min_x \left(f_1(x) - \frac{\alpha x}{2} - C_\alpha \right) = \frac{\alpha^2}{12},$$

$$\min_x \left(f_2(x) - \frac{\alpha x}{2} - C_\alpha \right) = \frac{(2-\alpha)^2}{48},$$

and

$$\min_x \left(f_3(x) - \frac{\alpha x}{2} - C_\alpha \right) = \begin{cases} \alpha(8-5\alpha)/48, & \alpha \le 4/3, \\ (2-\alpha)^2/12, & \alpha \ge 4/3. \end{cases}$$

The third is dominated by the lower envelope of the first two. Restoring the exact $-p/2$ contribution in each pointwise estimate and averaging produces the $-p/4$ term in Theorem 4.2.

Appendix B. Divisibility Correction

Let $P = x_1 \cdots x_p$ be a path and let $O \subseteq V(P)$ have even cardinality. Define

$$J = \{x_j x_{j+1} : |O \cap \{x_1, \ldots, x_j\}| \text{ is odd}\}.$$

Every internal vertex x_j changes prefix parity precisely when $x_j \in O$. Therefore

$$\mathrm{Odd}(J) = O.$$

Also $|E(J)| \leq p - 1$ and $\Delta(J) \leq 2$.

After parity correction, the minimum degree has fallen by at most two. In the sparse-independent regime it is still greater than $3p/4$ for all sufficiently large p, so Turán's theorem supplies a K_5. Deleting a C_4 inside that K_5 changes the edge count by 1 (mod 3), while deleting a C_5 changes it by 2 (mod 3). Every affected degree changes by two, so parity is preserved. The total degree loss from the path subgraph and the correcting cycle is at most four. Thus, when $q = o(p)$, the final graph has minimum degree at least $p - 1 - q - 4$, which eventually exceeds $(0.9 + \varepsilon_0)p$ for any fixed $0 < \varepsilon_0 < 0.1$.

Appendix C. Computational evidence

The computational audit descriptions and tested ranges are recorded in the separate formalization and reproducibility document. Their original appendix label is retained there. None is a premise of the mathematical argument.

Appendix D. Self-Contained List Edge Coloring

This appendix proves the list edge coloring theorem used in Section 7.2 (stated as Theorem 2.2), so that the corridor argument depends on no external coloring theorem. The result is the maximum-degree case of Galvin's theorem [10]; the gain graph of Section 7.2 is simple and bipartite, which is the case proved here.

D.1 Kernels

A **kernel** of a digraph D is an independent set K of vertices such that every vertex outside K has an out-neighbor in K. D is **kernel-perfect** if every induced subdigraph has a kernel. A kernel of a nonempty digraph is nonempty.

Lemma D.1 — Kernel coloring lemma

Let D be a kernel-perfect digraph in which every vertex v has a list $L(v)$ with $|L(v)| \geq d_D^+(v)+1$. Then the underlying graph of D has a proper coloring choosing each color from the corresponding list.

Proof

Pick any color c appearing in some list, let $S = \{v : c \in L(v)\}$, and let K be a kernel of $D[S]$. Color every vertex of K with c; this is proper on K because K is independent. Delete K from D and delete c from the list of every vertex of $S \setminus K$. The remaining digraph is an induced subdigraph of D, and hence is kernel-perfect. Every $v \in S \setminus K$ lost one color but also at least one out-neighbor, namely its out-neighbor in K, so the hypothesis $|L(v)| \geq d^+(v) + 1$ is preserved; vertices outside S keep their lists while their out-degrees do not increase. Repeat. Every uncolored vertex always has a nonempty list, and each round with $S \neq \varnothing$ colors the nonempty set K, so every vertex is eventually colored. □

D.2 Stable pairings

Let B be a bipartite graph with parts U and R in which every vertex z has a linear preference order $>_z$ on its incident edges. A matching $M \subseteq E(B)$ is **stable** if every edge $f \notin M$ has an endpoint z that is covered by an edge $e \in M$ with $e >_z f$.

Lemma D.2 — Gale–Shapley [9]

For every system of preferences, a stable matching exists.

Proof

Run deferred acceptance. While some $u \in U$ is unmatched and has not yet proposed along all of its edges, let u propose along its most preferred untried edge, say f with endpoint $r \in R$. If r is unmatched, or prefers f to its currently held edge, then r accepts f, releasing its previous partner, if any; otherwise r rejects f. Every edge is proposed at most once, so the process terminates, and it terminates in a matching M.

Let $f = ur \notin M$. Suppose first that u proposed along f at some point. Then r either rejected f immediately or accepted it and later released it; in both cases, at that moment r held or received an edge it strictly prefers to f. Since the edge held by r only improves during the process, the final edge $e \in M$ at r satisfies $e >_r f$. Suppose instead that u never proposed along f. An unmatched vertex of U keeps proposing while untried edges remain, so at termination u is matched, say to $e \in M$; and since u proposes in decreasing order of preference and never reached f, we have $e >_u f$. In both cases f has an endpoint whose matched edge is preferred to f. □

D.3 The coloring theorem

Theorem D.3 — Galvin, maximum-degree case

Let B be a simple bipartite graph with maximum degree Δ, and let every edge e carry a list $L(e)$ with $|L(e)| \geq \Delta$. Then B has a proper edge coloring choosing each color from the corresponding list.

Proof

Step 1: a proper Δ-edge-coloring. By König's edge coloring theorem [13], B has a proper edge coloring $\varphi : E(B) \to \{1, \ldots, \Delta\}$. For completeness: induct on the number of edges among bipartite graphs of maximum degree at most Δ. Remove an edge $e = ur$ and color the rest; the induction hypothesis applies because the remaining graph still has maximum degree at most Δ. At most $\Delta - 1$ colors appear at u and at most $\Delta - 1$ at r, so u misses a color α and r misses a color β. If they miss a common color, give it to e. Otherwise the subgraph formed by the α-

and β-colored edges has maximum degree at most two, and u has degree one in it. Hence the alternating component starting at u is a well-defined simple path P, whose edges are colored alternately β, α. The path P cannot end at r: since r misses β, such a path would end with an α-edge and hence have even length, while every path between the adjacent vertices u and r of a bipartite graph has odd length. Swap the colors α and β along P; by maximality of P the coloring stays proper, the vertex r is untouched, and u now misses β. Give β to e.

Step 2: the orientation. Define preferences from φ: every $u \in U$ prefers its incident edges with **higher** φ-color, every $r \in R$ prefers **lower** φ-color. Define a digraph D on the vertex set $E(B)$: for distinct edges e, f sharing an endpoint z, put an arc $e \to f$ exactly when $f >_z e$.

Out-degrees: let $e = ur$ with $\varphi(e) = c$. The out-neighbors of e are the edges at u with color greater than c — at most $\Delta - c$, since the colors at u are pairwise distinct — and the edges at r with color smaller than c — at most $c - 1$. Hence

$$d_D^+(e) \le (\Delta - c) + (c - 1) = \Delta - 1.$$

Step 3: kernel-perfectness. An induced subdigraph of D is $D[S]$ for a set S of edges of B. A kernel of $D[S]$ is exactly a stable matching of the subgraph $(V(B), S)$ with the inherited preferences: independence in D means that no two kernel edges share an endpoint, because two edges at a common vertex are always joined by an arc; and the domination condition asks that every $f \in S \setminus K$ have an arc into K, that is, an endpoint z and an edge $e \in K$ at z with $e >_z f$, which is stability. Lemma D.2 provides a stable matching, so D is kernel-perfect.

Step 4: conclusion. Every edge has $|L(e)| \ge \Delta \ge d_D^+(e) + 1$, so Lemma D.1 yields a proper coloring of the underlying graph of D from the lists — that is, a proper list edge coloring of B. □

Remark D.4 — Application

The gain graph of Section 7.2 is simple and bipartite, and hypothesis (7.2) bounds every list by $b - t_i \ge \max\{\rho, u\} \ge \Delta$ of the gain graph. Theorem D.3 therefore applies verbatim. The multigraph version of Galvin's theorem holds as well [10] and is not needed here.

Appendix E. Further consequences of the constructions

This appendix is a curated selection rather than a declaration inventory. It retains results that sharpen the paper's main constructions, give a reusable mathematical interface, or record a nontrivial exact form. Encoding lemmas, monotonicity wrappers, routine arithmetic conversions, and duplicate certificates remain in the Lean sources. None of the statements below is an additional premise of Theorem 1.1.

In E.1–E.2, $p, q, n = p + q$, and d_i have their meanings from Section 2; $\nu_3'(G)$ is the split-graph packing observable used by the Lean development, and $\mathrm{rp}(p)$ is its round-robin factor count. In E.2 we write $q = 2p - s$, as in Section 5.

E.1 Factorization packing bounds

Corollary E.1.1 — Single factor-assignment packing

For every split graph G and every injection

$$\sigma : \mathrm{Fin}(\mathrm{rp}(p)) \hookrightarrow \mathrm{Fin}(q),$$

the triangle packing obtained by assigning the round-robin factors of the clique to distinct independent vertices satisfies

$$\sum_j |F_j(\sigma(j))| \le \nu_3'(G).$$

Here $F_j(i)$ is the set of clique edges in the j-th factor that can be completed to triangles through independent vertex i. Distinct factors use distinct hosts, and each factor is a matching, so the resulting triangles are edge-disjoint.

Corollary E.1.2 — Double-factorization lower bound

If $\mathrm{rp}(p) \le q$ and $q \ge 2$, then

$$\frac{1}{q}\sum_i \binom{d_i}{2} + \frac{\mathrm{doubledFactors}}{\mathrm{rp}(p)} \cdot \frac{\mathrm{dispersionV}}{q(q-1)} \le \nu_3'(G).$$

The first term is the mean clique-edge coverage obtained by averaging factor assignments. The second is the dispersion gain from compatible double use of a factor.

Corollary E.1.3 — Reserved-gain packing bound

Let $R \subseteq [p]$ be a reserved clique set and $J \subseteq [q]$ a set of independent hosts. Under the host-size and round-robin feasibility hypotheses of Section 7,

$$\binom{p-|R|}{2} + \binom{|R|}{2} - \frac{(2(p-|R|)-1)A_J - A_{2J}}{2|J|} + (1-\theta_R)\frac{(|J| - \mathrm{rp}(p-|R|))B_J}{|J|} \le \nu_3'(G).$$

The quantities $A_J, A_{2J}, B_J, \theta_R$ are the shifted-center parameters defined in Section 7.

E.2 Effective corridor interfaces

Corollary E.2.1 — Short-corridor constant $3/2$

If $p \ge 37$ and $s^2 \le 36p$, then

$$\frac{p}{2} + \frac{s^2 - 6s + 3}{12} \le \frac{3}{2}(3p - s).$$

This is the arithmetic estimate that supplies the explicit $3/2$ constant in the short corridor.

Corollary E.2.2 — Packing-to-partition bridge

For every split graph,

$$\mathrm{cp}(G) \le \Phi(G) = |E(G)| - 2\nu_3(G).$$

Consequently, under the low-corridor hypotheses $p \ge 37$ and $s^2 \le 36p$,

$$\mathrm{cp}(G) \le \frac{n^2}{6} + \frac{3}{2}n.$$

The mid-corridor and high-ratio specializations are already stated in Proposition 10.5 and are not repeated here.

E.3 Exact common-profile family $H(p, q, d)$

Put

$$A = \frac{C_2(p) + qd}{3}, \qquad B = C_2(d) + C_2(p-d), \qquad C = C_2(d) + \frac{d(p-d) + C_2(p-d)}{3},$$

and $F(p, q, d) = \min\{A, B, C\}$.

Corollary E.3.1 — Exact fractional optimum

For $p \ge 3$, $q \ge 1$, and $d \le p$, finite packing/covering LP duality and Theorem 3.1 give

$$\tau_3^*(H(p, q, d)) = \nu_3^*(H(p, q, d)) = F(p, q, d).$$

Corollary E.3.2 — Phase regions

The three branches have the following certified sufficient regions:

- if $3d^2 - 3dp - dq + p^2 - p \ge 0$ and $q + 1 \le d$, then $\tau_3^*(H) = A$;
- if $B \le A$ and $p \le 2d + 1$, then $\tau_3^*(H) = B$;
- if $C \le A$ and $2d + 1 \le p$, then $\tau_3^*(H) = C$.

These statements package the branch comparisons used in the exact LP analysis without restating every pairwise polynomial inequality.

E.4 Unified fractional margin

Corollary E.4.1 — Fractional slack lower bound

For split graphs with $p \ge 3$ and $1 \le q \le 2p$,

$$T(G) + \mu(\alpha)p^2 - \frac{p}{4} \le \tau_3^*(G).$$

Here $T(G)$, $\mu(\alpha)$, and α are the threshold and profile parameters of Section 4. The estimate records the quantitative slack available before the integral rounding step.

E.5 Complete-split sharpness package

Let $S_p = K_p \vee \overline{K}_{2p}$, so $|V(S_p)| = 3p$.

Corollary E.5.1 — Exact value and normal form

For every $p \geq 1$,

$$\mathrm{cp}(S_p) = 2p^2 - \binom{p}{2} = \frac{|V(S_p)|^2}{6} + \frac{|V(S_p)|}{6}.$$

This is Corollary 10.2a in its integer and quadratic-plus-linear forms.

Corollary E.5.2 — Forced linear coefficient

For every $C \in \mathbb{R}$, if

$$\mathrm{cp}(G) \leq \frac{n^2}{6} + Cn$$

holds for every split graph G, then $C \geq 1/6$.

E.6 Reusable formalized components

This section highlights components with uses beyond the complete-split application. Their possible upstream destination is separate from the mathematical claims of Paper III.

E.6.1 Finite packing/covering LP duality

For every finite 0/1 incidence system, the maximum fractional packing value equals the minimum fractional cover value. The proof is implemented through a finite conic-Farkas argument.

E.6.2 Dirac-type matching and dense correction

Under the Dirac minimum-degree hypothesis, an even finite graph has a perfect matching, with a corresponding near-perfect form in odd order. In the sparse-independent regime these matching tools feed a correction deleting at most $p + 8$ edges, with incidence loss at most 6, so that the residual graph has even degrees and edge count divisible by 3.

E.6.3 Uniform nibble interface

Assume the formalized majority-nibble interface Nibble.NibbleTheoremMost. Let $r \geq 2$. For every $\beta > 0$, it supplies parameters $\mu, \eta, d_0 > 0$ such that every r-uniform hypergraph satisfying the majority near-regularity and codegree hypotheses at scale $d \geq d_0$ has a matching M with

$$|M| \geq (1 - \beta)\frac{|W|}{r}.$$

E.7 Packing witnesses and the edge-gap form

Corollary E.7.1 — Attainment and fractional edge bound

For every finite graph G, the integral optimum is attained by an edge-disjoint triangle packing M, and

$$\nu_3^*(G) \leq \frac{|E(G)|}{3}.$$

Corollary E.7.2 — Majority-nibble edge-gap form

For every $\beta > 0$, under the majority near-regularity and codegree hypotheses supplied by NibbleTheoremMost,

$$\nu_3^*(G) - \nu_3(G) \le \frac{\beta |E(G)|}{3}.$$

The routine inequalities $\nu_3 \le \nu_3^*$, $|E| \le |V|^2/2$, their monotonicity wrappers, and duplicate vertex-normalized forms are omitted from this appendix.

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